

Setting up Development Environments

Setting up a Web-based Development Environment w/ Colab

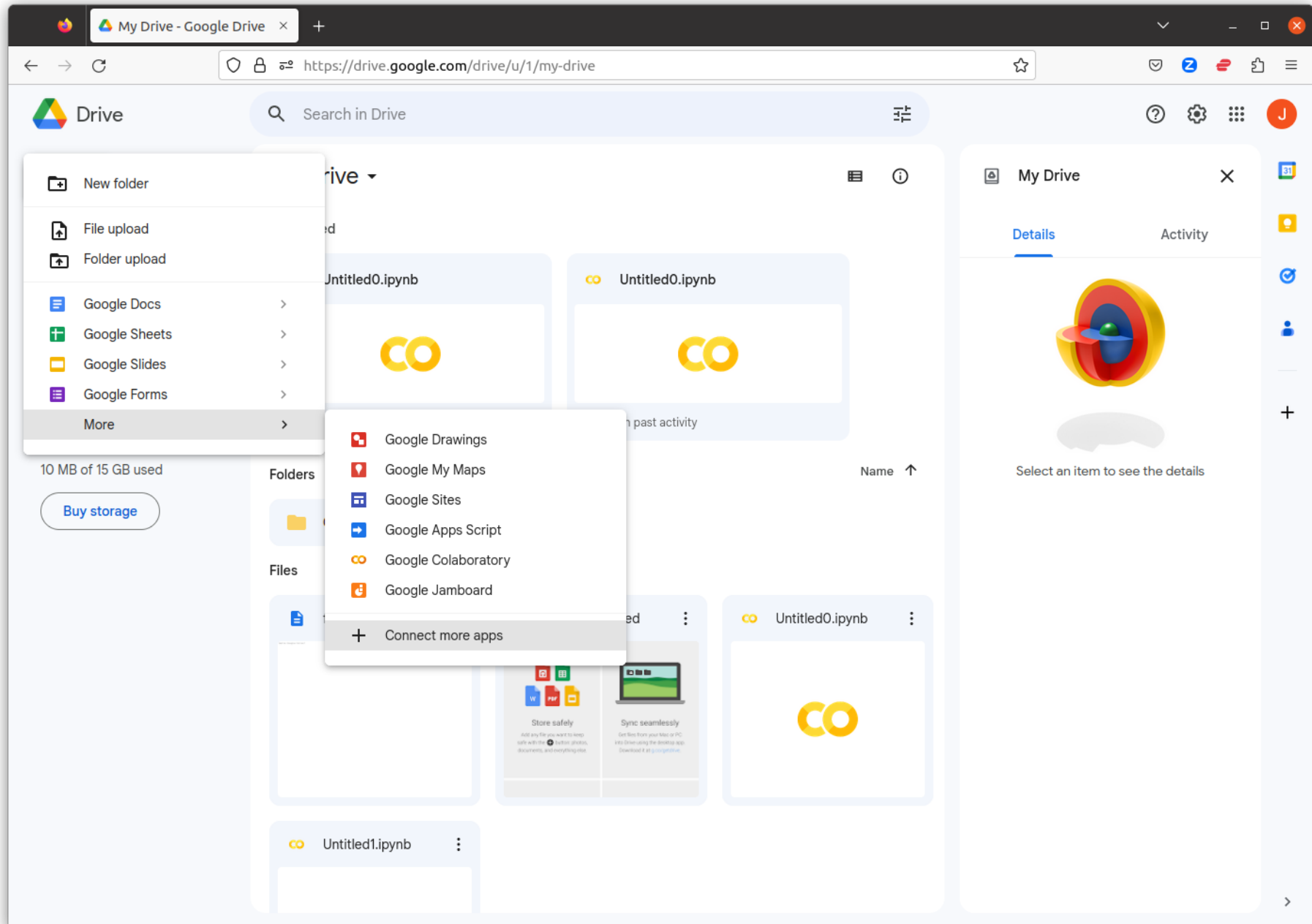
Basic process, reviewed in subsequent slides

1. go to google drive
2. connect colaboratory app
3. create a notebook w/ colaboratory
4. edit notebook settings to enable GPU acceleration
5. mount a google drive

Note: If you do not have the Colaboratory app connected, you can add it by following these steps:

1. Open Google Drive
2. Click on the "New" button
3. Select "More" and then "Connect more apps."
4. Search for "Colaboratory" and click "Connect."

Colaboratory in Google Drive - Connect More Apps



Colaboratory in Google Drive - Search Store

The screenshot shows a web browser window with the Google Drive interface. The address bar displays `https://drive.google.com/drive/u/1/my-drive`. The left sidebar shows the Drive navigation menu with options like 'New', 'My Drive', 'Computers', 'Shared with me', 'Recent', 'Starred', 'Trash', and 'Storage'. The main area displays the Google Workspace Marketplace search results for 'Colaboratory'. The search bar at the top of the marketplace shows 'Colaboratory' with a search icon and a close button. Below the search bar, there are filters for 'All Filters', 'Works with', and 'Price'. The search results are titled 'Search results for Colaboratory' and include a disclaimer: 'Google doesn't verify reviews or ratings. [Learn more about reviews and results](#)'. The results are displayed as a grid of application cards. The first card is for 'Colaboratory' by the 'Colaboratory team', which is marked as 'Installed'. It has a 4.7 star rating and over 10,000,000 downloads. The description states: 'This allows Google Colaboratory to open and create files in Google Drive. It i...'. Other visible cards include 'MAIL MERGE for SHEETS' by yamm, 'Form Publisher' by Talarian, 'GET DATA INTO SHEETS' by Talarian, 'Awesome Table - Data ...' by Talarian, 'AdvancedSummary', 'Dr Merge: Free Unlimite...', 'YHFINANCE', and 'MarketView - By Wealth...'. The bottom of the screen shows a snippet of a Jupyter Notebook file named 'Untitled1.ipynb'.

My Drive - Google Drive

Search in Drive

Google Workspace Marketplace

Colaboratory

All Filters Works with Price

Search results for Colaboratory

Google doesn't verify reviews or ratings. [Learn more about reviews and results](#)

Colaboratory
Colaboratory team
This allows Google Colaboratory to open and create files in Google Drive. It i...
★ 4.7 * 10,000,000+

MAIL MERGE for SHEETS
yamm
Yet Another Mail Merge:...
Talarian
Send mass personalized emails with your Gmail™ account. Track engagement of each email in...
★ 4.7 * 10,000,000+

Form Publisher
Talarian
Generate templated documents with Google Forms. Generate PDF, Google Docs™, Sheets™, ...
★ 4.6 * 10,000,000+

GET DATA INTO SHEETS
Talarian
Export data from Salesforce, Quickbooks, and many other sources to Google Sheets™, N...
★ 4.5 * 10,000,000+

Awesome Table - Data ...
Talarian
Export data from Salesforce, Quickbooks, and many other sources to Google Sheets™, N...
★ 4.5 * 10,000,000+

AdvancedSummary
Advanced Summary by ...

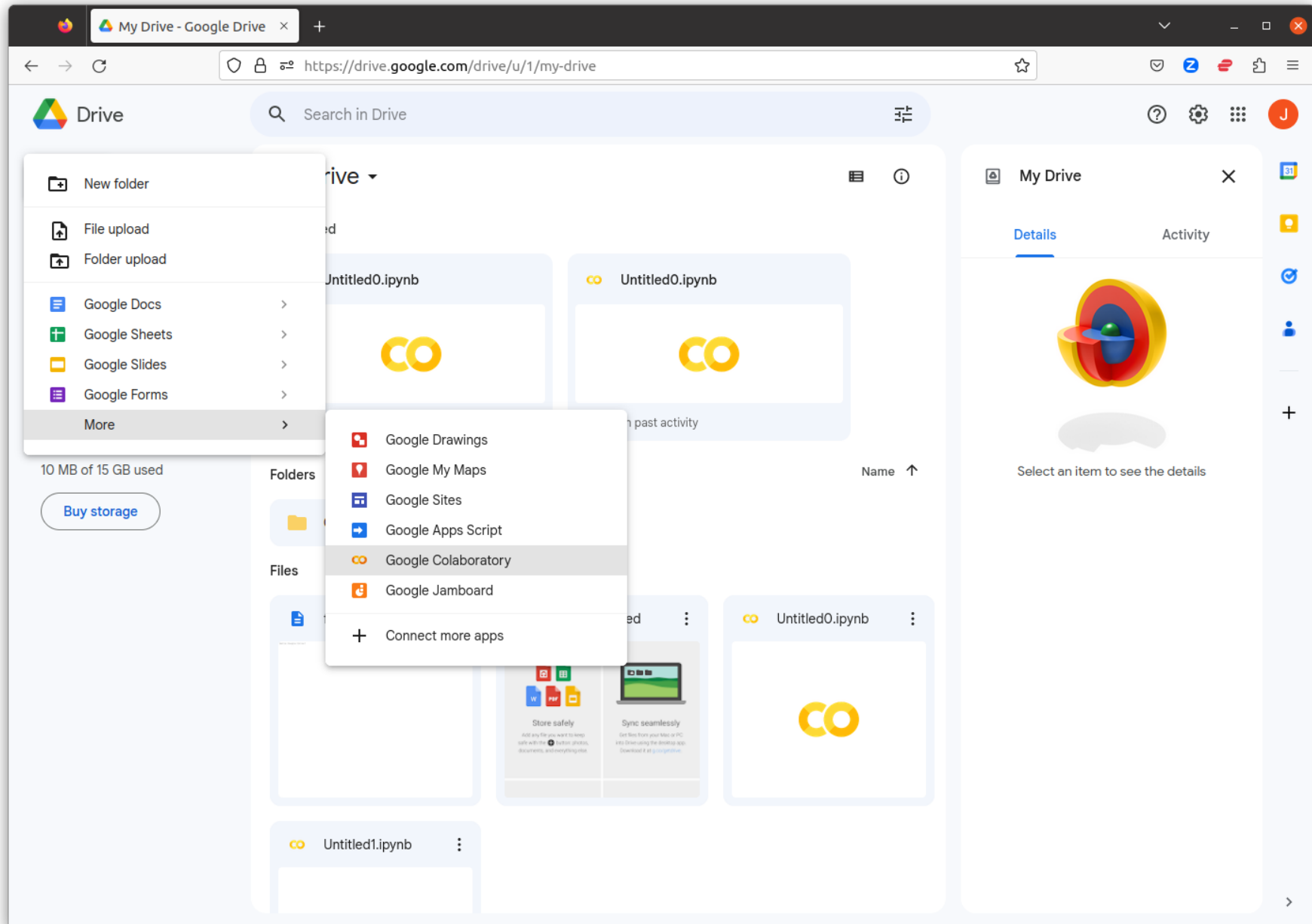
Dr Merge: Free Unlimite...
Dr Merge: Free Unlimite...

YHFINANCE
YHFINANCE

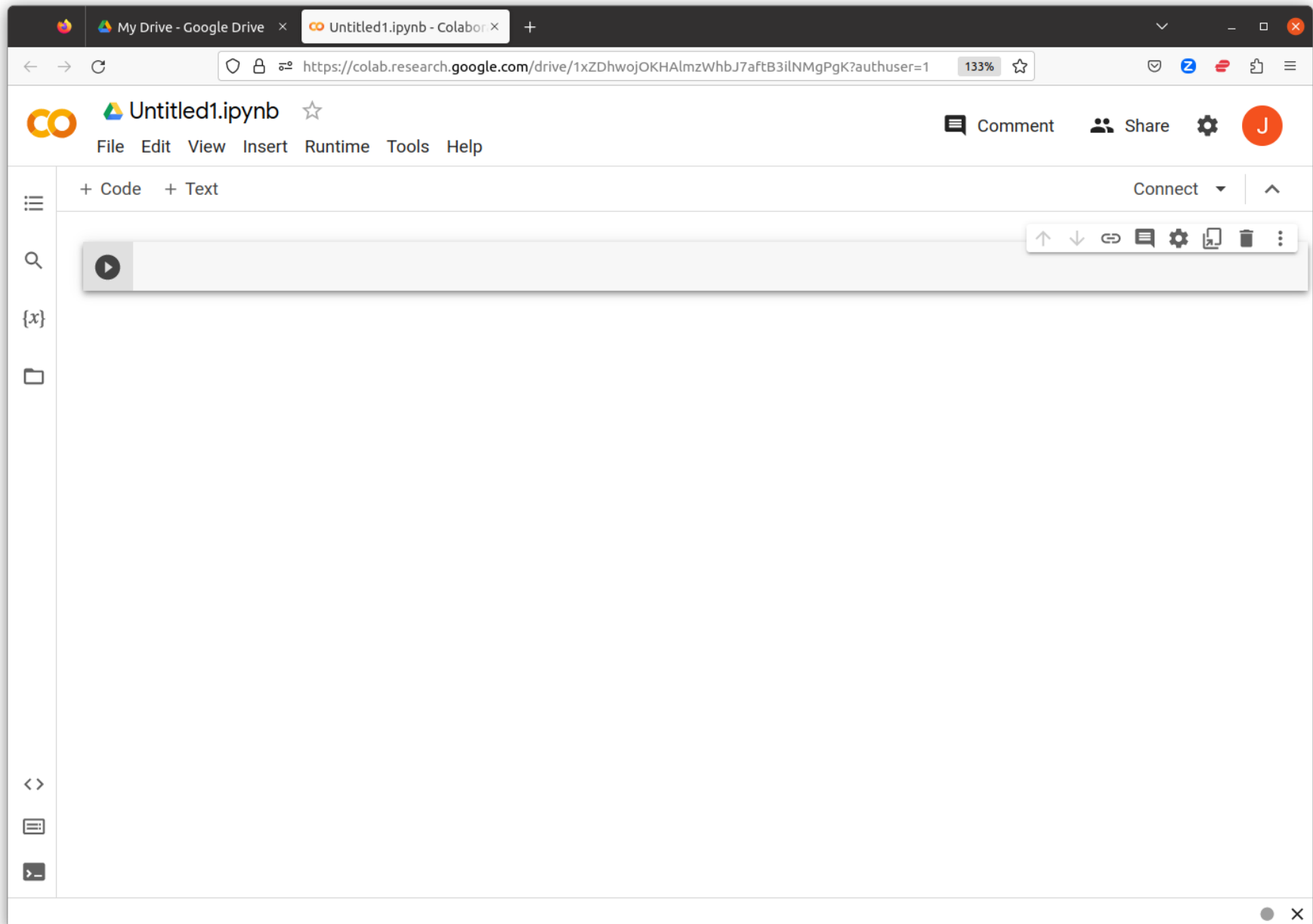
WEALTH ASSISTANTS
MarketView - By Wealth...

Untitled1.ipynb

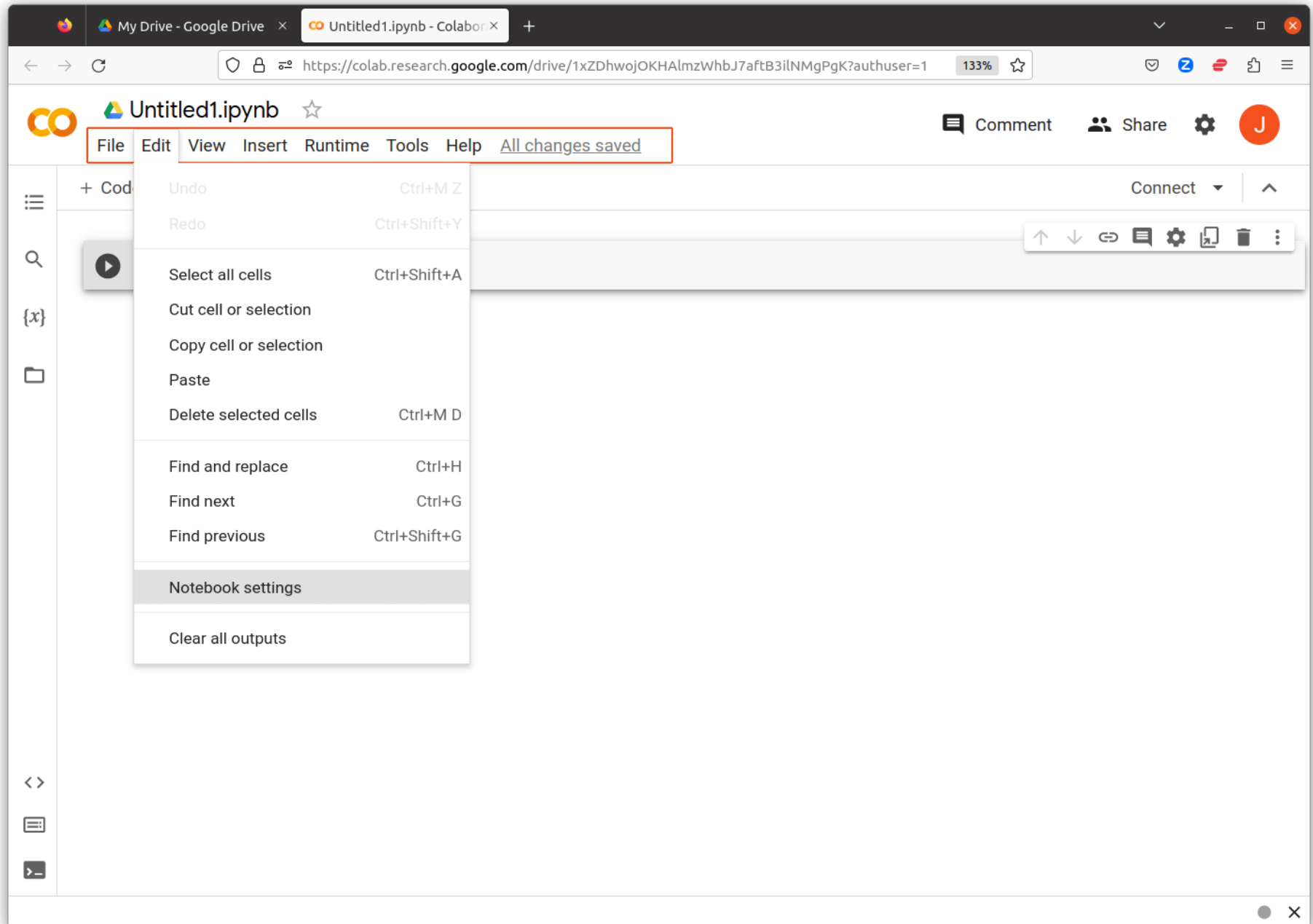
Colaboratory in Google Drive - Open The Notebook



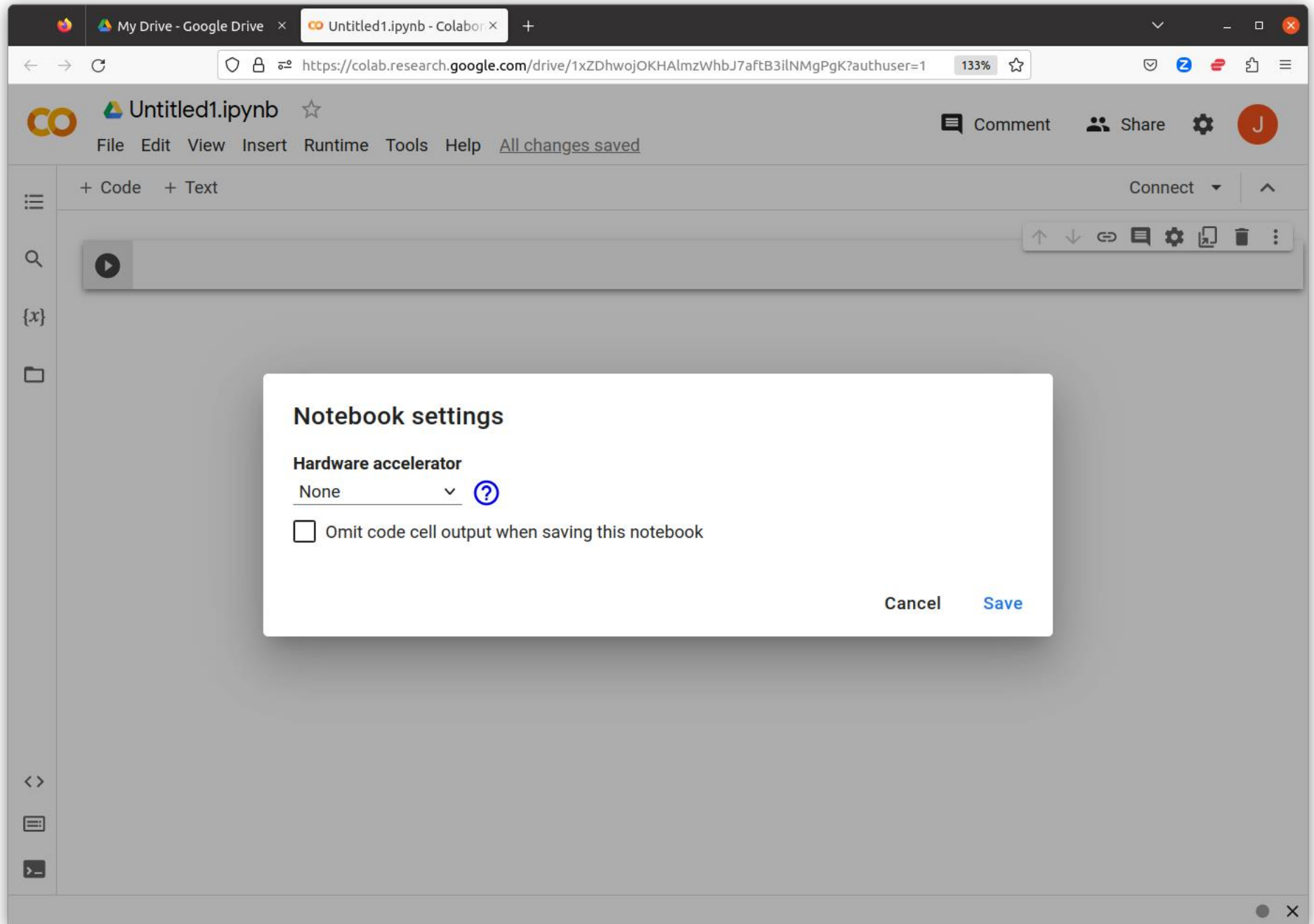
Colaboratory in Google Drive - Notebook is Open



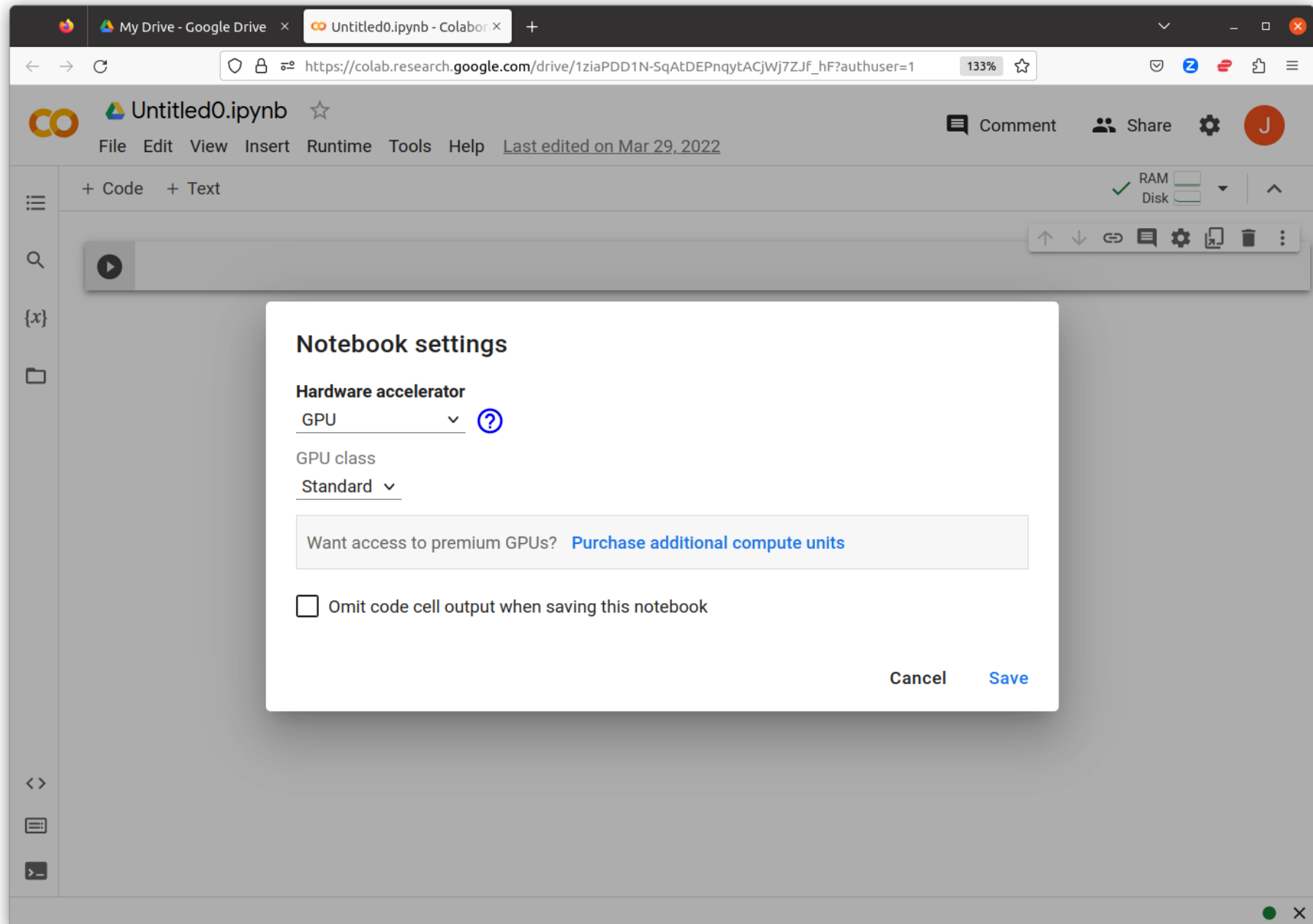
Colaboratory in Google Drive - Edit Notebook Setting



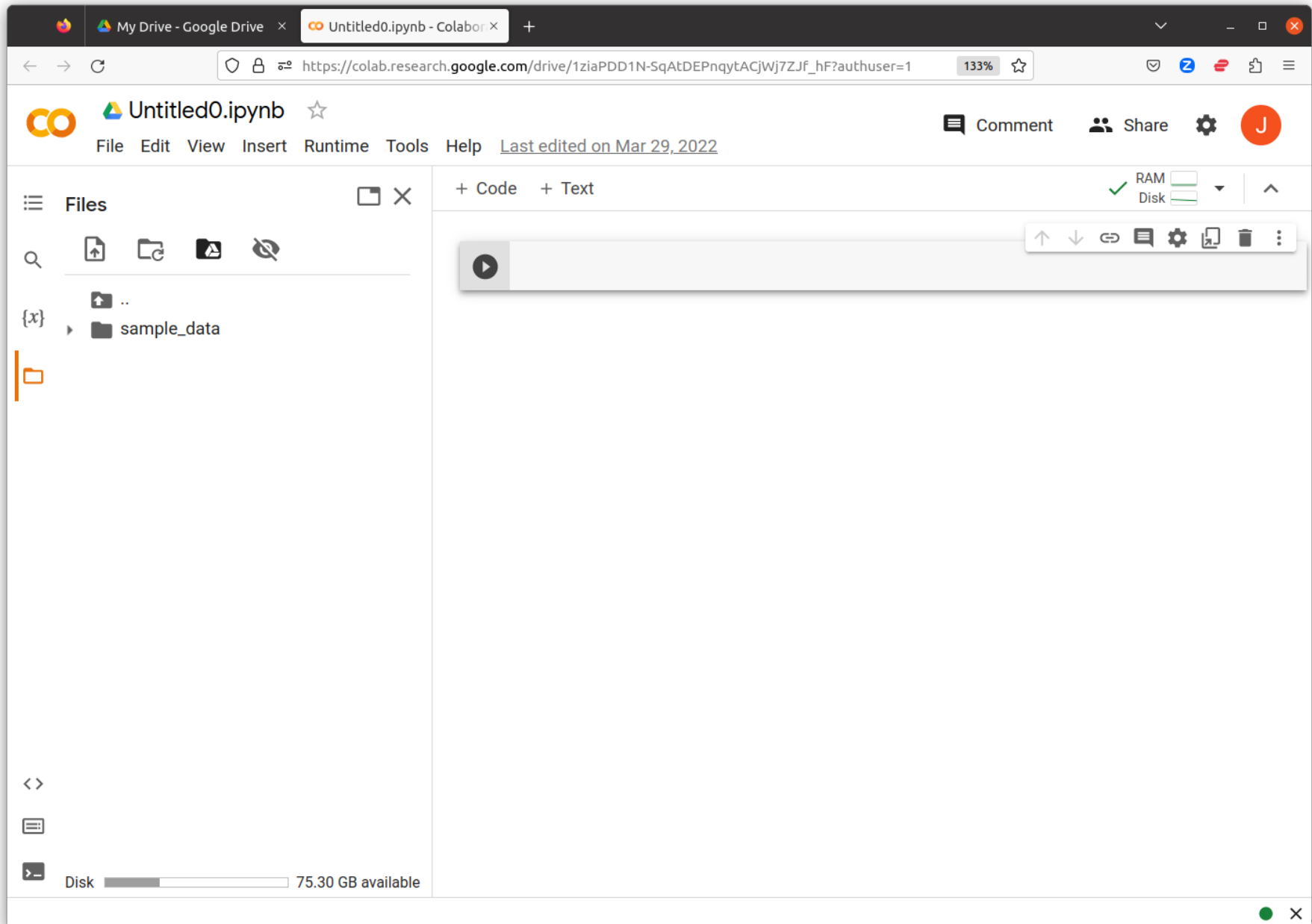
Colaboratory in Google Drive - Select GPU



Colaboratory in Google Drive - GPU is Selected



Colaboratory in Google Drive - Mount Drive



Mount Google Drive in Colab

In your Colab notebook, run the following code to mount your Google Drive:

```
from google.colab import drive  
drive.mount('/content/drive')
```

Follow the authorization steps to give Colab access to your Google Drive.

Save a file to Google Drive:

```
import pandas as pd
```

```
df = pd.DataFrame({'Column1': [1, 2, 3], 'Column2': [4, 5, 6]}) # Example: Save a DataFrame as a CSV
```

```
df.to_csv('/content/drive/My Drive/filename.csv', index=False)
```

Download Directly to Your Local Machine

```
from google.colab import files
```

```
df.to_csv('filename.csv', index=False) # Save a file to local disk
```

```
files.download('filename.csv') # Download the file
```

Setting up a Local Development Environment

1. (optional - only if you have nvidia GPU) install recent driver for your GPU
 - test with `nvidia-smi`
2. if windows: install windows subsystem for linux 2 and ubuntu.
3. install miniconda <https://docs.conda.io/projects/miniconda/en/latest/>
(using conda envs enables 1 system to have isolated pythons of different versions)
4. create a conda environment w/ python 3.9–3.11, e.g.

```
conda create -n gradAppML2024 python=3.10
```

5. activate the created conda environment

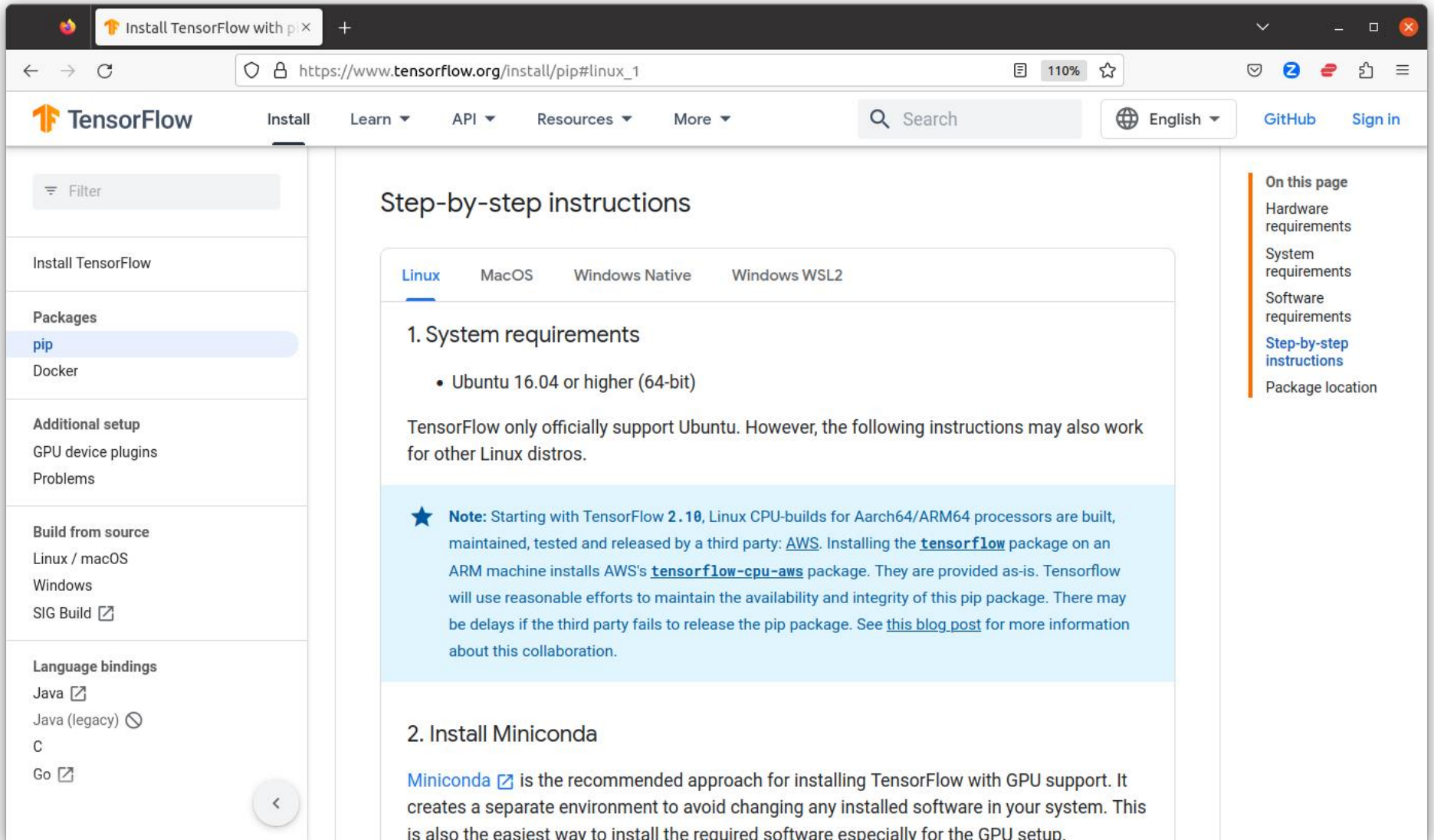
```
conda activate gradAppML2024
```

do this every time you start, can deactivate when done

6. In setup conda env, follow guidelines at <https://www.tensorflow.org/install> and <https://www.tensorflow.org/install/pip> closely to ease uniformity.
 - scroll down to “step-by-step” instead of quick version at top.
 - select your OS then follow instructions
 - NB: if a M1 or M2 mac, when following https://www.tensorflow.org/install/pip#macos_1 DON'T overlook the footnote! In particular, for M1 or M2 (non-intel) macs, you need to use <https://developer.apple.com/metal/tensorflow-plugin/> instead.

Setting up a Local Development Environment

- Select your OS (ubuntu)



The screenshot shows the TensorFlow website's installation page for Linux using pip. The browser address bar shows the URL https://www.tensorflow.org/install/pip#linux_1. The page has a navigation bar with links for Install, Learn, API, Resources, and More, along with a search bar and language selector (English). A left sidebar contains a 'Filter' section and a list of categories: Install TensorFlow, Packages (with 'pip' selected), Docker, Additional setup (GPU device plugins, Problems), Build from source (Linux / macOS, Windows, SIG Build), and Language bindings (Java, Java (legacy), C, Go). The main content area is titled 'Step-by-step instructions' and has tabs for Linux, MacOS, Windows Native, and Windows WSL2. The 'Linux' tab is active, showing '1. System requirements' which lists 'Ubuntu 16.04 or higher (64-bit)'. It also states that TensorFlow officially supports Ubuntu but provides instructions for other Linux distros. A blue note box mentions that starting with TensorFlow 2.10, Linux CPU-builds for Aarch64/ARM64 processors are built, maintained, tested, and released by a third party: AWS. It advises installing the `tensorflow` package on an ARM machine, which will install AWS's `tensorflow-cpu-aws` package. It notes that TensorFlow will use reasonable efforts to maintain the availability and integrity of this pip package, but there may be delays if the third party fails to release the pip package, and refers to a blog post for more information. Below this, '2. Install Miniconda' is shown, recommending Miniconda as the best approach for installing TensorFlow with GPU support, as it creates a separate environment to avoid changing any installed software in the system and is the easiest way to install the required software, especially for the GPU setup.

TensorFlow

Install Learn API Resources More

Search English GitHub Sign in

Filter

Install TensorFlow

Packages

pip

Docker

Additional setup

GPU device plugins

Problems

Build from source

Linux / macOS

Windows

SIG Build

Language bindings

Java

Java (legacy)

C

Go

Step-by-step instructions

Linux MacOS Windows Native Windows WSL2

1. System requirements

- Ubuntu 16.04 or higher (64-bit)

TensorFlow only officially support Ubuntu. However, the following instructions may also work for other Linux distros.

★ **Note:** Starting with TensorFlow 2.10, Linux CPU-builds for Aarch64/ARM64 processors are built, maintained, tested and released by a third party: [AWS](#). Installing the `tensorflow` package on an ARM machine installs AWS's `tensorflow-cpu-aws` package. They are provided as-is. Tensorflow will use reasonable efforts to maintain the availability and integrity of this pip package. There may be delays if the third party fails to release the pip package. See [this blog post](#) for more information about this collaboration.

2. Install Miniconda

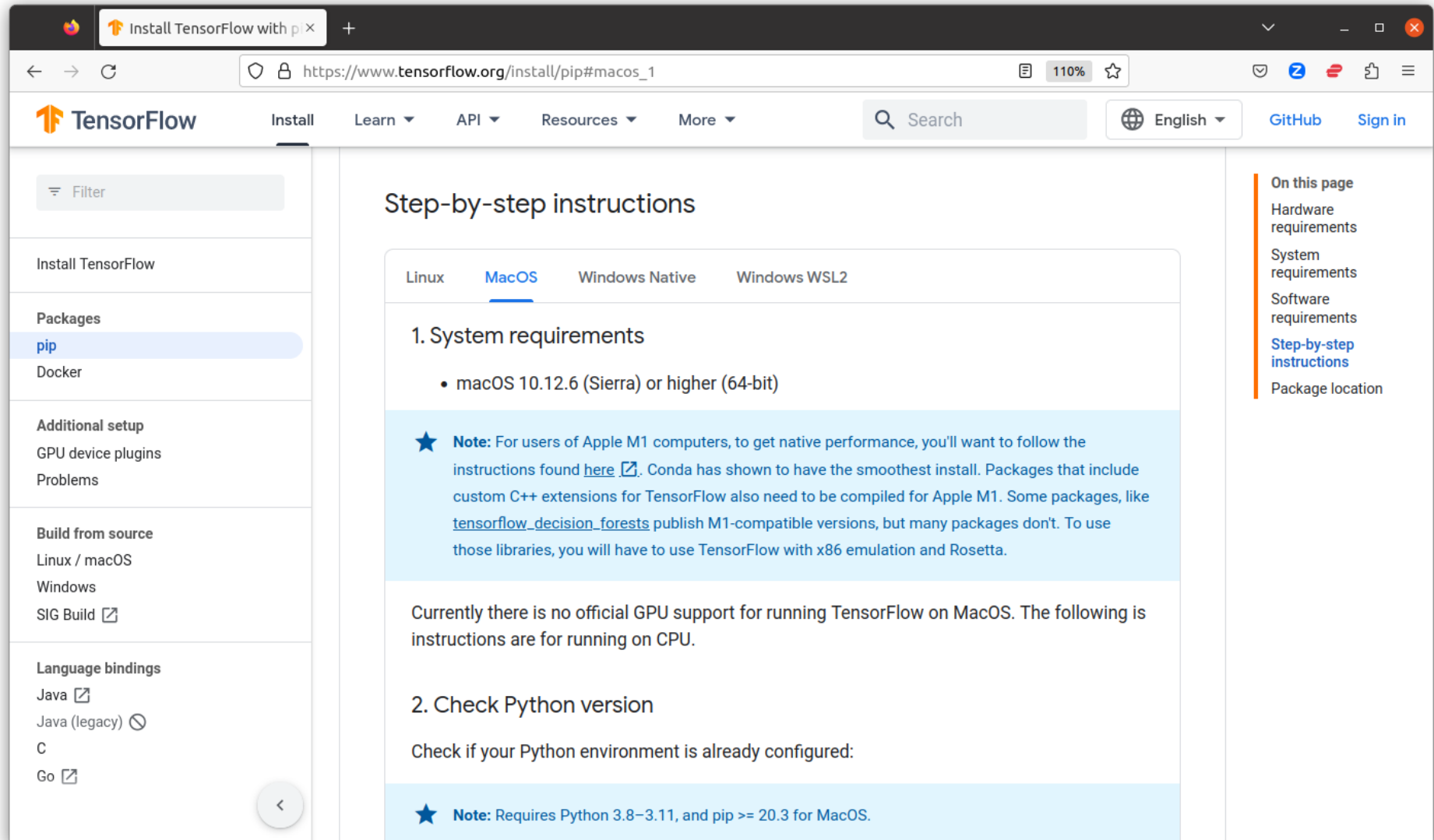
[Miniconda](#) is the recommended approach for installing TensorFlow with GPU support. It creates a separate environment to avoid changing any installed software in your system. This is also the easiest way to install the required software especially for the GPU setup.

On this page

- Hardware requirements
- System requirements
- Software requirements
- Step-by-step instructions
- Package location

Setting up a Local Development Environment

- Select your OS (MacOs)



The screenshot shows the TensorFlow website's installation page for macOS. The browser address bar displays `https://www.tensorflow.org/install/pip#macos_1`. The page features a sidebar on the left with navigation links: 'Install TensorFlow', 'Packages' (with 'pip' selected), 'Additional setup', 'Build from source', and 'Language bindings'. The main content area is titled 'Step-by-step instructions' and has tabs for 'Linux', 'MacOS' (selected), 'Windows Native', and 'Windows WSL2'. Under the 'MacOS' tab, the first section is '1. System requirements', which lists 'macOS 10.12.6 (Sierra) or higher (64-bit)'. A blue note box highlights that for Apple M1 computers, native performance requires following specific instructions, mentioning Conda and M1-compatible packages. Below this, it states that there is no official GPU support for TensorFlow on macOS, with instructions provided for CPU. The second section is '2. Check Python version', which includes a note requiring Python 3.8–3.11 and pip >= 20.3 for macOS. A right-hand sidebar lists 'On this page' links: 'Hardware requirements', 'System requirements', 'Software requirements', 'Step-by-step instructions' (highlighted), and 'Package location'.

TensorFlow

Install TensorFlow with pip

https://www.tensorflow.org/install/pip#macos_1

TensorFlow

Install Learn API Resources More

Search

English

GitHub Sign in

Filter

Install TensorFlow

Packages

pip

Docker

Additional setup

GPU device plugins

Problems

Build from source

Linux / macOS

Windows

SIG Build

Language bindings

Java

Java (legacy)

C

Go

Step-by-step instructions

Linux MacOS Windows Native Windows WSL2

1. System requirements

- macOS 10.12.6 (Sierra) or higher (64-bit)

Note: For users of Apple M1 computers, to get native performance, you'll want to follow the instructions found [here](#). Conda has shown to have the smoothest install. Packages that include custom C++ extensions for TensorFlow also need to be compiled for Apple M1. Some packages, like [tensorflow_decision_forests](#) publish M1-compatible versions, but many packages don't. To use those libraries, you will have to use TensorFlow with x86 emulation and Rosetta.

Currently there is no official GPU support for running TensorFlow on MacOS. The following is instructions are for running on CPU.

2. Check Python version

Check if your Python environment is already configured:

Note: Requires Python 3.8–3.11, and pip >= 20.3 for MacOS.

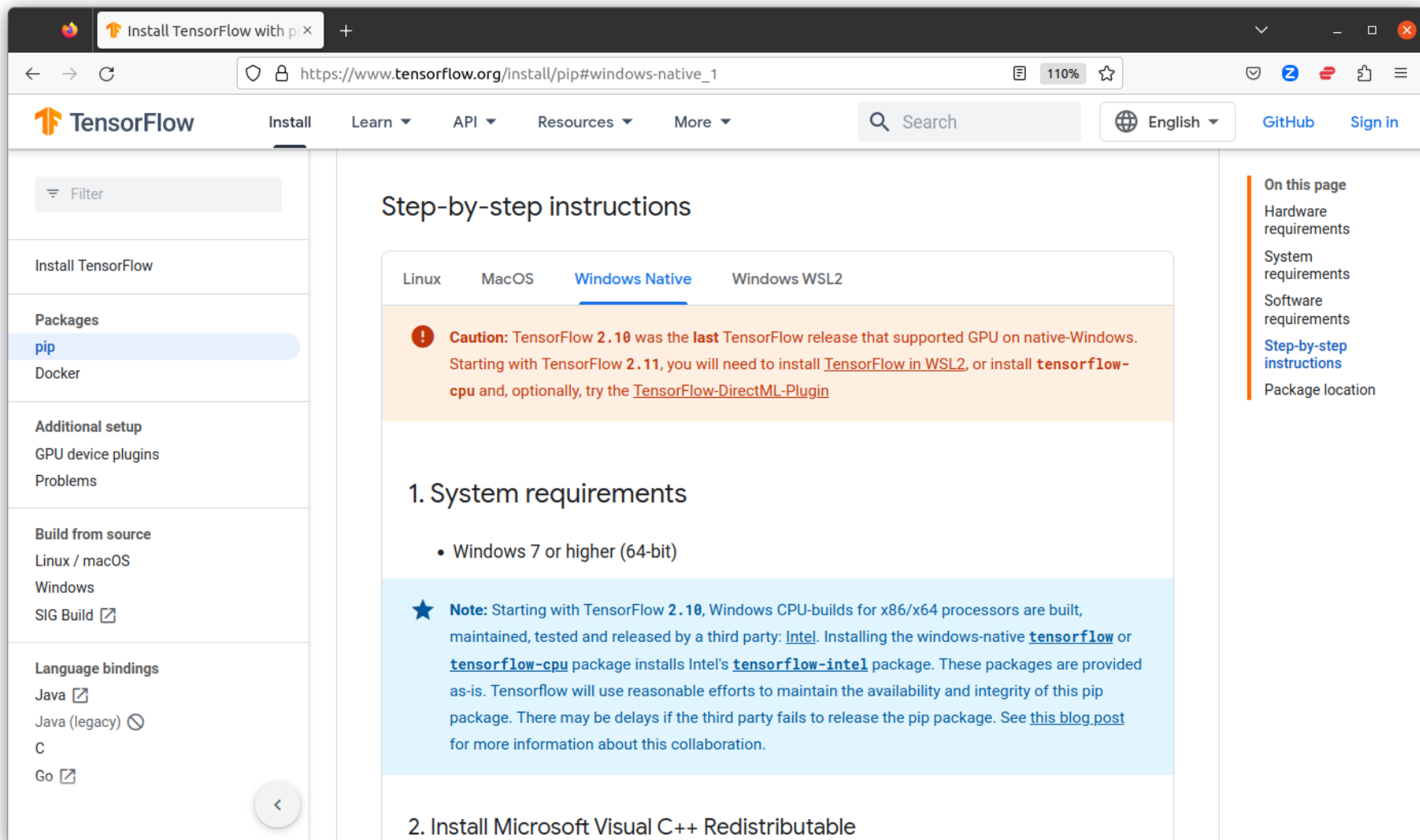
On this page

- Hardware requirements
- System requirements
- Software requirements
- Step-by-step instructions
- Package location

Setting up a Local Development Environment

- Select your OS (Windows)

Both native windows and windows subsystem for linux (recommended) available.



The screenshot shows the TensorFlow website's installation page for pip on Windows Native. The page is titled "Step-by-step instructions" and has tabs for Linux, MacOS, Windows Native (selected), and Windows WSL2. A caution box states: "Caution: TensorFlow 2.10 was the last TensorFlow release that supported GPU on native-Windows. Starting with TensorFlow 2.11, you will need to install TensorFlow in WSL2, or install tensorflow-cpu and, optionally, try the TensorFlow-DirectML-Plugin". The main content is divided into two sections: "1. System requirements" and "2. Install Microsoft Visual C++ Redistributable". The first section lists "Windows 7 or higher (64-bit)". A note box states: "Note: Starting with TensorFlow 2.10, Windows CPU-builds for x86/x64 processors are built, maintained, tested and released by a third party: Intel. Installing the windows-native tensorflow or tensorflow-cpu package installs Intel's tensorflow-intel package. These packages are provided as-is. Tensorflow will use reasonable efforts to maintain the availability and integrity of this pip package. There may be delays if the third party fails to release the pip package. See this blog post for more information about this collaboration." The left sidebar contains a navigation menu with categories: "Install TensorFlow", "Packages" (with "pip" selected), "Additional setup", "Build from source", and "Language bindings". The right sidebar contains a "On this page" section with links to "Hardware requirements", "System requirements", "Software requirements", "Step-by-step instructions" (selected), and "Package location".

TensorFlow

Install Learn API Resources More

Search

English

GitHub Sign in

Filter

Install TensorFlow

Packages

pip

Docker

Additional setup

GPU device plugins

Problems

Build from source

Linux / macOS

Windows

SIG Build

Language bindings

Java

Java (legacy)

C

Go

Step-by-step instructions

Linux MacOS **Windows Native** Windows WSL2

Caution: TensorFlow 2.10 was the last TensorFlow release that supported GPU on native-Windows. Starting with TensorFlow 2.11, you will need to install [TensorFlow in WSL2](#), or install **tensorflow-cpu** and, optionally, try the [TensorFlow-DirectML-Plugin](#)

1. System requirements

- Windows 7 or higher (64-bit)

Note: Starting with TensorFlow 2.10, Windows CPU-builds for x86/x64 processors are built, maintained, tested and released by a third party: [Intel](#). Installing the windows-native **tensorflow** or **tensorflow-cpu** package installs Intel's **tensorflow-intel** package. These packages are provided as-is. Tensorflow will use reasonable efforts to maintain the availability and integrity of this pip package. There may be delays if the third party fails to release the pip package. See [this blog post](#) for more information about this collaboration.

2. Install Microsoft Visual C++ Redistributable

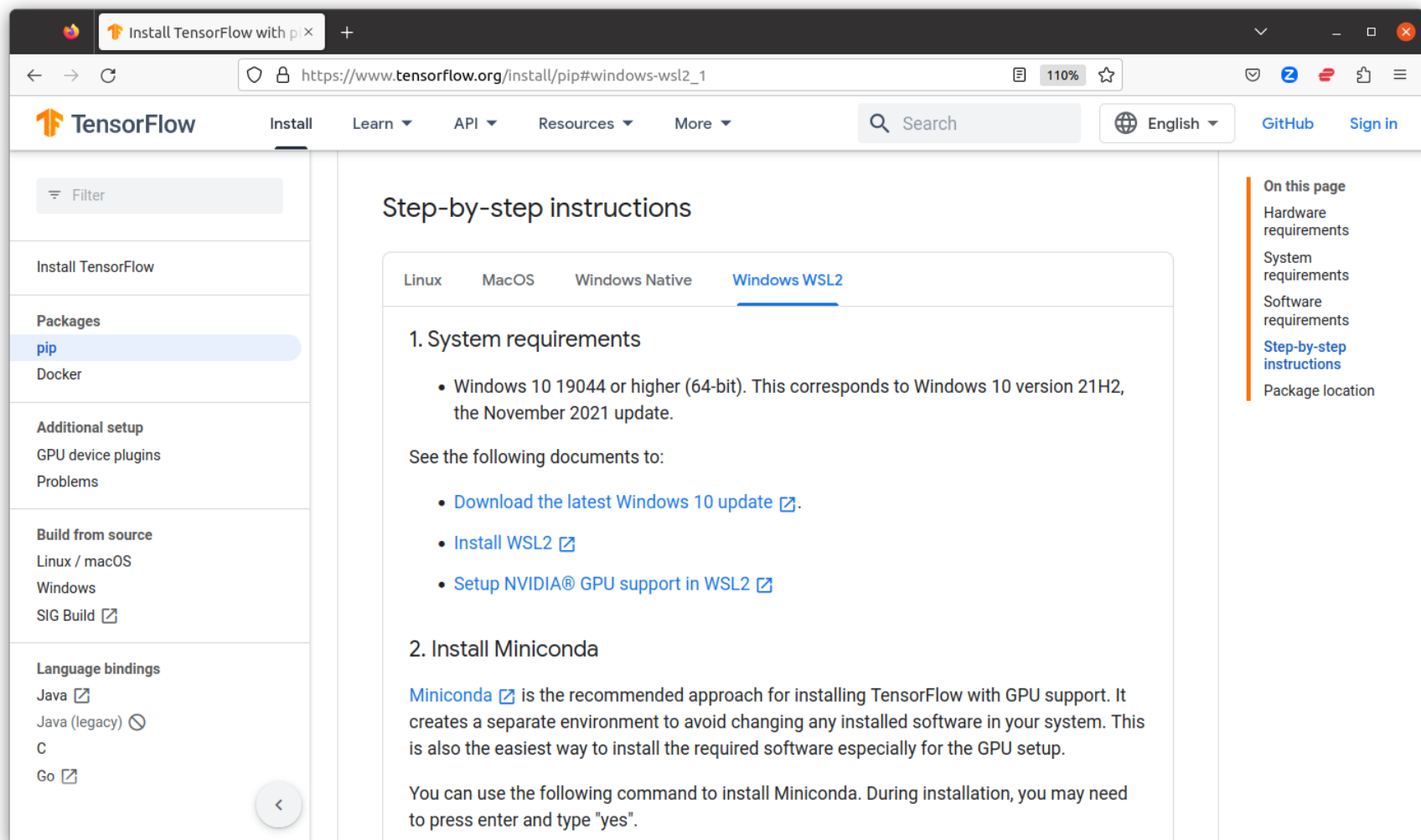
On this page

- Hardware requirements
- System requirements
- Software requirements
- Step-by-step instructions**
- Package location

Setting up a Local Development Environment

- Select your OS (WSL-ubuntu)

For WSL, have to first set up HyperV and WSL (involve enabling options in Win 11), then running installation like ubuntu.



The screenshot shows the TensorFlow website's installation page for Windows WSL2. The browser address bar displays `https://www.tensorflow.org/install/pip#windows-wsl2_1`. The page features a left sidebar with navigation links: 'Install TensorFlow', 'Packages' (with 'pip' selected), 'Additional setup', 'Build from source', and 'Language bindings'. The main content area is titled 'Step-by-step instructions' and has tabs for 'Linux', 'MacOS', 'Windows Native', and 'Windows WSL2'. Under '1. System requirements', it lists: 'Windows 10 19044 or higher (64-bit). This corresponds to Windows 10 version 21H2, the November 2021 update.' and 'See the following documents to:'. Below this are links to 'Download the latest Windows 10 update', 'Install WSL2', and 'Setup NVIDIA® GPU support in WSL2'. Under '2. Install Miniconda', it states: 'Miniconda is the recommended approach for installing TensorFlow with GPU support. It creates a separate environment to avoid changing any installed software in your system. This is also the easiest way to install the required software especially for the GPU setup.' and provides a command to install Miniconda. A right sidebar lists 'On this page' with links to 'Hardware requirements', 'System requirements', 'Software requirements', 'Step-by-step instructions' (highlighted), and 'Package location'.

TensorFlow

Install Learn API Resources More

Search

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Build from source

Linux / macOS

Windows

SIG Build

Language bindings

Java

Java (legacy)

C

Go

Step-by-step instructions

Linux MacOS Windows Native **Windows WSL2**

1. System requirements

- Windows 10 19044 or higher (64-bit). This corresponds to Windows 10 version 21H2, the November 2021 update.

See the following documents to:

- [Download the latest Windows 10 update](#)
- [Install WSL2](#)
- [Setup NVIDIA® GPU support in WSL2](#)

2. Install Miniconda

[Miniconda](#) is the recommended approach for installing TensorFlow with GPU support. It creates a separate environment to avoid changing any installed software in your system. This is also the easiest way to install the required software especially for the GPU setup.

You can use the following command to install Miniconda. During installation, you may need to press enter and type "yes".

On this page

- Hardware requirements
- System requirements
- Software requirements
- Step-by-step instructions**
- Package location

7. making sure your python is from the conda env you made on WSL/linux/macOS:

`which python`

on windows

`where python`

8. Upgrade pip (python package manager)

`python -m pip install --upgrade pip`

9. `pip install tensorflow`

. if you have nvidia GPU use instead

`pip install tensorflow[and-cuda]`

10. Install jupyter, scikit learn, numpy, matplotlib

`python -m pip install -U jupyter matplotlib numpy pandas scipy scikit-learn`

11. start jupyter notebook server & open browser and connect to server

`jupyter notebook`

12. install the gcloud CLI following instructions for your OS at

<https://cloud.google.com/sdk/docs/install>.

Google Cloud Platform

1. 2-5 projects require GPUs, but they are unnecessary (and unhelpful) for 1.
2. For 3-5th project have applied for 50 USD in Google Cloud Platform education credits for each of you to get you state of the art GPU access.
3. Click a link I'll post in course website
4. Enter your abc123@drexel.edu email address.
5. A confirmation will be sent to you with a coupon code.
6. redeem the coupon code
7. follow instructions I will post to request a quota increase on GPUs.
8. Can either use web interface or command line to start a VM in the cloud.
9. Will share slides/instructions reviewing how to kick off and interact with a jupyter notebook server running on a VM instance w/ GPU.

Setting up a Local Development Environment

- mini-assignment

Homework/Assignment - try to complete before next Monday, Set up a local installation of miniconda, make an environment w/ appropriate python, install tensorflow following the tensorflow installation instructions on the web (they are summarized on the previous slide). Upload screenshots of the results of the two commands listed under "verify install".

Also, check that your jupyter installation worked correctly, and include a screenshot of running jupyter.

Also, show a screenshot of installed gcloud CLI.

Get help at office hours on Wednesday or Friday if you encounter issues

Textbook's Github Repo

- <https://homl.info> redirects to
- <https://github.com/ageron/handson-ml3/>
- for previous version (only needed for reinforcement learning lectures/assignment)
- <https://github.com/ageron/handson-ml2/>